

Samvaad-Setu: Detailed Solution Document

A multilingual, confidence-aware voice assistant for Karnataka's 1092 citizen helpline

1) Executive Overview

Samvaad-Setu is designed as a real-time voice pipeline that prioritizes one core principle: **correct understanding before action**. Instead of directly acting on first-pass NLP output, the system asks the citizen to verify the AI's interpretation, then decides whether to continue, clarify, or escalate to a human agent based on confidence and sentiment signals.

Primary outcomes expected from this design:

- Reduced misrouting of grievances through explicit verification.
- Better handling of dialect variation across Karnataka districts.
- Safer response behavior under distress and uncertainty.
- Operational auditability through persistent event logs and feedback loops.

2) Problem Context and Why This Matters

Citizen helpline calls often involve noisy audio, regional dialects, code-mixed language, and emotionally charged scenarios. A straightforward speech-to-intent pipeline is not enough because errors at early stages cascade into incorrect downstream actions. Samvaad-Setu addresses this with verification-first logic, confidence scoring, and escalation safeguards.

3) System Architecture

- **Frontend:** Flutter Web app with citizen view and agent dashboard.
- **Backend:** FastAPI with WebSocket-first orchestration for low-latency turn processing.
- **State:** Redis for ephemeral session state and quick retrieval.
- **Durability:** Postgres for audit logs and verified interaction datasets.
- **AI services:** Sarvam AI for ASR/TTS, Gemini for NLU, librosa for prosodic features.

4) End-to-End Turn Pipeline

- 1 **ASR:** Speech is transcribed with confidence signals.
- 2 **Prosody extraction:** Pitch, energy, and speaking-rate cues are computed.
- 3 **Dialect-aware NLU:** District profile and taxonomy constraints guide intent extraction.
- 4 **Sentiment fusion:** Text and voice stress cues are combined into intensity + label.
- 5 **Verification prompt:** AI rephrases the interpreted issue and asks for confirmation.
- 6 **Citizen response branch:** correct / partial / incorrect handling updates state.
- 7 **Composite confidence scoring:** uncertainty and distress become explicit decision signals.

- 8 **Escalation decision:** continue, clarify again, or handoff to human dashboard.
- 9 **TTS:** assistant response delivered in language/dialect register.
- 10 **Audit + feedback write:** key events logged for traceability and model improvement.

5) Verification Engine (Core Differentiator)

The verification engine enforces explicit citizen confirmation before intent commitment. Its three-state response model reduces silent AI failure modes:

- **correct:** mark session as confirmed and proceed to next decision layer.
- **partial:** increment clarification counter, ask focused clarifying question.
- **incorrect:** re-ask if attempts remain; escalate after repeated mismatch.

This mechanism directly aligns with public-service reliability requirements where wrong categorization can delay resolution for citizens.

6) Dialect Context and Taxonomy Control

District-level dialect profiles and a fixed grievance taxonomy constrain LLM ambiguity. Instead of free-form intent labels, the model selects from policy-aligned categories (for example, water supply, ration card status, BBMP services, BESCOM outages, and distress emergency).

7) Confidence and Escalation Logic

- Composite score combines ASR confidence, intent entropy, sentiment intensity, and clarification count.
- Low composite confidence triggers additional clarification.
- Very low confidence triggers escalation with reason: `low_confidence`.
- High distress sentiment triggers escalation with reason: `high_distress`.
- Repeated clarification failures trigger escalation with reason: `repeated_clarification`.

8) Agent Dashboard Support

- Queue endpoint ranks escalations by urgency (sentiment + age).
- Full-context endpoint provides transcript, confidence history, and audit summary.
- Real-time WebSocket updates notify connected agents of new escalations.
- Agent correction endpoint updates audit log and feedback corpus.

9) Privacy, Compliance, and Trust

- PII redaction before LLM calls (phone, Aadhaar-like numbers, names, address patterns).
- Audio retention disabled by default and controlled with explicit flags.
- Immutable audit events capture every state transition and actor action.
- Feedback exports use structured JSONL for controlled model-retraining pipelines.

10) Deployment and Operations

- One-command startup with Docker Compose (backend + frontend + Redis + Postgres).
- Health endpoints for service checks and rolling latency metrics (p50/p95).
- Idempotent session creation to handle duplicate client retries safely.
- Graceful fallback paths for ASR and TTS service outages.

11) Demo Strategy (Judge-Friendly)

- Scenario A: Mangaluru Kannada query that demonstrates verification loop behavior.
- Scenario B: Distressed Hindi caller that triggers automatic escalation.
- Scenario C: Low-quality/noisy audio showing confidence drop and safe handling.
- Agent dashboard handoff proof with editable fields and audit trace visibility.

12) Risks, Mitigations, and Next Steps

- **Risk:** dialect coverage gaps. **Mitigation:** expand district profiles and add agent corrections to retraining loop.
- **Risk:** model drift or API instability. **Mitigation:** fallback models + cached classifiers + observable health metrics.
- **Risk:** false sentiment spikes. **Mitigation:** multi-modal fusion, thresholds, and human override.
- **Next step:** integrate telephony channels, expand district taxonomy, and strengthen offline resiliency.

Conclusion

Samvaad-Setu provides a practical and safety-aware architecture for multilingual public-service voice support. Its verification-first approach, confidence-aware branching, and audit-ready data model make it suitable for high-stakes grievance handling where correctness, transparency, and human escalation are essential.